

Karla Madrigal

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PROFESSIONAL SUMMARY

Driven Electrical Engineering student with a strong interest in aerospace and RF technologies, motivated by their high-impact applications and complex engineering outcomes. Experienced in embedded systems, PCB design, firmware development, and delivering scalable engineering solutions for technical teams with a focus on project delivery and systems integration.

EDUCATION

Cullen College of Engineering, University of Houston | Houston, TX
Bachelor of Science in Electrical Engineering; Junior

Expected Graduation Date: May 2029

PROJECTS

Custom Flight Computer - Avionics & Telemetry Board

June 2026 – Present

- Designing and assembling a high-reliability flight computer incorporating an STM32 MCU, IMU, barometer, and mLRS (MavLink LoRa) transceiver for long-range telemetry transmission.
- Developing real-time embedded C++ firmware for sensor data logging, state estimation, radio communications, and parachute deployment logic.
- Designing a dedicated calibration board to measure S-parameters using a VNA to optimize RF performance prior to launch.

Micromouse - Autonomous Maze Solving Robot

Fall 2024 – May 2025

- Designed a compact, 2-layer, SAMD21-based PCB, integrating a motor driver, IMU, boost converter, and IR sensors, optimizing autorouter performance and minimizing trace length.
 - Calibrated 7 IR sensors for obstacle avoidance and debugged I2C communication protocols.
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EXPERIENCE

Create@UH Hardware Design Lab Assistant | Cullen College of Engineering, Houston, TX

August 2025 – Ongoing

- Maintained safety protocols, emphasizing protective measures such as power tool risk mitigation using PPE during lab safety onboarding training for students.
- Coordinated and executed the first soldering workshop within the Create@UH program, collaborating with Cullen College of Engineering Student Organizations to strengthen student engagement.

Amazon Future Engineer Software Engineering Intern | Amazon, Bellevue, WA

Summer 2025, Summer 2026

Amazon Supply Chain Optimization Technologies | **Summer 2026**

- Designed and implemented an end-to-end data ingestion and agentic analytics pipeline using AWS Lambda, DynamoDB, and LLM orchestration to streamline log parsing and anomaly detection for on-call engineers.
- Deployed the automated diagnosis workflow to production, significantly reducing incident investigation time from several days to minutes while maintaining high data accuracy.

Amazon Fulfillment Technologies | **Summer 2025**

- Built a full-stack internal web app with React.js, TypeScript, AWS Lambda, CDK, and DynamoDB for warehouse staff to update operational parameters, saving hours of engineering support time.
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LEADERSHIP

Institute of Electrical and Electronics Engineers – University of Houston Student Branch

Fall 2024 – Spring 2026

- **Vice President** | December 2025 – May 2026
 - Established and strengthened our connection with IEEE Houston Section, IEEE Region 5, and related professional networks.
 - **Project and Workshops Chair** | August 2025 – December 2025
 - Mentored committee members to design, debug, and teach analog circuits, enabling a rotating peer-led workshop model.
 - Hosted workshops teaching practical engineering skills (Git, EasyEDA, KiCad, and Soldering) to 50+ members.
 - Held weekly office hours to support students in circuit design, debugging, and programming.
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SKILLS AND CERTIFICATES

Embedded & Hardware: SolidWorks, KiCad, LTspice | **Software:** C/C++, PlatformIO, Git, MATLAB